

Question Bank:

1. Find the Fourier transform of the function, $f(t) = \begin{cases} 0 & \text{for } -\infty \leq t \leq a \\ t & \text{for } a < t \leq b \\ 0 & \text{for } t > b \end{cases}$
 Ans: $F(\omega) = \frac{1}{i\omega} (ae^{-i\omega a} - be^{-i\omega b}) + \frac{1}{\omega^2} (e^{-i\omega b} - e^{-i\omega a})$
2. Find Fourier transform of $f(t) = \begin{cases} \sin t & \text{for } 0 < t < \pi \\ 0 & \text{otherwise} \end{cases}$
 Ans: $F(\omega) = \frac{(1 + e^{-i\omega\pi})}{1 - \omega^2}$
3. Given $F(e^{-t^2}) = \sqrt{\pi} e^{-\frac{\omega^2}{4}}$, find the Fourier transform of i) $e^{-\frac{t^2}{3}}$ ii) $e^{-4(t-3)^2}$
 Ans: $\sqrt{\pi} e^{-\frac{3\omega^2}{4}}$; $\sqrt{\pi} e^{-\omega(3i + \frac{\omega^2}{16})}$
4. Find the Fourier transform of the function,
 $f(t) = \begin{cases} a^2 - t^2 & \text{for } |t| \leq a \\ 0 & \text{for } |t| > a \end{cases}$
 Ans: $F(\omega) = \frac{4}{\omega^3} (\sin a\omega - a\omega \cos a\omega)$
5. Find the Fourier sine and cosine transform of $f(t) = \begin{cases} k & \text{if } 0 < t < a \\ 0 & \text{otherwise} \end{cases}$
 Ans: $F_s(\omega) = \frac{k}{\omega} (1 - \cos a\omega)$; $F_c(\omega) = \frac{k}{\omega} \sin a\omega$
6. Find the Fourier sine transform of $f(t) = \frac{1}{t(1+t^2)}, t > 0$.
 Ans: $\sqrt{\frac{\pi}{2}} (1 - e^{-\omega})$
7. Find the Fourier sine transform of $f(t) = \frac{1}{t}, t > 0$.
 Ans: $\frac{\pi}{2}$
8. Find inverse Fourier sine transform of $\frac{e^{-a\omega}}{a}, a > 0$.
 Ans: $\frac{2}{\pi} \tan^{-1} \left(\frac{\omega}{a} \right)$
9. Find the Fourier cosine transform of $f(x) = e^{-ax}, a > 0$. Hence find $\int_0^\infty \frac{\cos \omega x d\omega}{(1+\omega^2)}$
 Ans: $F_c(\omega) = \frac{a}{a^2 + \omega^2}$; $\int_0^\infty \frac{\cos \omega x d\omega}{(1+\omega^2)} = \frac{\pi}{2a} e^{-ax}$
10. Find the Fourier cosine transform of $f(t) = e^{-at} \cos at$.
 Ans: $F_c(\omega) = \frac{a(2a^2 + \omega^2)}{(2a^2 - \omega^2)^2}$

11. Find the Fourier sine and cosine transform of $f(t) = \begin{cases} \cos t & \text{if } 0 < t < a \\ 0 & \text{otherwise} \end{cases}$

$$\text{Ans: } F_c(\omega) = \frac{\{\omega \sin \omega a \cos a - \cos \omega a \sin a\}}{\omega^2 - 1}$$

12. Find $f(t)$ satisfying the integral equation, $\int_0^\infty f(t) \cos \omega t dt = e^{-\omega}$

$$\text{Ans: } f(t) = \frac{2}{\pi(1 + \omega^2)}$$
